FOR A GLUEING PAIR THE FIRST TOR OF THE BIGGER RING WITH THE COKERNEL OF LOCALIZING A MODULE AT THE ELEMENT VANISHES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Glueing pairs (R -> R', f) and the modules R/R[f^infty] and R'/R'[f^infty]. the setting at statement line 25503 and used at proof lines 25518-25520. Fixes the objects appearing in the special case and the comparison M tensor_R R' = R'/R'[f^infty].
- Commutation of Tor with filtered colimits. invoked at proof lines 25513-25514. Reduces the statement for M_f/M to the statements for the quotients M/f^nM.
- The long exact sequence for Tor and the injectivity of the multiplication map in the displayed diagram. used at proof lines 25521-25536. Is the computation that kills Tor_1 in the special case.
- Exactness of tensoring with R' for the quotients involved. invoked at proof lines 25545-25546. Lets the general case be reduced to the free case.

Source reference. Original lemma 00M0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 806–806. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0BNF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14663–14663. Cross-chapter navigation pointer into

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Source reference. Original lemma 25486: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25486–25486. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 25255: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25255–25255. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 24173: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24173–24173. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(R \to R', f)$ be a glueing pair. For every R-module M, we have $Tor_1^R(R', \mathrm{Coker}(M \to M_f)) = 0$.*

Proof. Set $\overline{M} = M/M[f^\infty]$. Then $\mathrm{Coker}(M \to M_f) \cong \mathrm{Coker}(\overline{M} \to \overline{M}_f)$ hence we may and do assume that f is a nonzerodivisor on M. In this case $M \subset M_f$ and $M_f/M = \mathrm{colim}\, M/f^nM$ where the transition maps are given by multiplication by f. Since formation of Tor groups commutes with colimits (Algebra, Lemma 0BNF) it suffices to show that $\mathrm{Tor}_1^R(R', M/f^nM) = 0$.

We first treat the case $M = R/R[f^\infty]$. By Lemma 25255 we have $M \otimes_R R' = R'/R'[f^\infty]$. From the short exact sequence $0 \to M \to M \to M/f^nM \to 0$ we obtain the exact sequence

$$\begin{array}{ccccc} \mathrm{Tor}_1^R(R', R/R[f^\infty]) & \longrightarrow & \mathrm{Tor}_1^R(R', M/f^nM) & \longrightarrow & R'/R'[f^\infty] \\ & & & \overset{f^n}{\searrow} & \\ R'/R'[f^\infty] & \longrightarrow & (R'/R'[f^\infty])/(f^n(R'/R'[f^\infty])) & \longrightarrow & 0 \end{array}$$

by Algebra, Lemma 00M0. Here the diagonal arrow is injective. Since the first group $\mathrm{Tor}_1^R(R', R/R[f^\infty])$ is zero by Lemma 25486, we deduce that $\mathrm{Tor}_1^R(R', M/f^nM) = 0$ as desired.

To treat the general case, choose a surjection $F \to M$ with F a free $R/R[f^\infty]$-module, and form an exact sequence

$$0 \to N \to F/f^nF \to M/f^nM \to 0.$$

By Lemma 24173 this sequence remains unchanged, and hence exact, upon tensoring with R'. Since $\mathrm{Tor}_1^R(R', F/f^nF) = 0$ by the previous paragraph, we deduce that $\mathrm{Tor}_1^R(R', M/f^nM) = 0$ as desired. □